



Cambridge IGCSE™

ENVIRONMENTAL MANAGEMENT

0680/21

Paper 2 Management in Context

October/November 2022

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)	272 899.8 / 272 900;	1
1(b)	<i>any three from:</i> hot / warm / high temperatures (for growth); high rainfall (for growth); (conditions suitable for) photosynthesis / stated equation / to produce glucose; irrigation / water can be stored;	3
1(c)	Malaga AND more or twice number of sunshine hours ; OR Santander AND rain all year round / Málaga has little rain in June and July ;	1
1(d)(i)	C ;	1
1(d)(ii)	E ;	1
1(d)(iii)	F AND low concentration or less of all three ions / not fertile / not suitable for agriculture;	1
1(d)(iv)	71;	1
1(d)(v)	160 ÷ 6 or 26.6 / 26.7 26.67; 27;	2
1(d)(vi)	fertiliser / organic matter / crop rotation / plant legumes;	1
1(d)(vii)	<i>ease of cultivation:</i> easy / quickly loses nutrients / low nutrient value; <i>drainage:</i> quick / fast / good / high / does not hold water; <i>air content:</i> high / good / holds a lot of air;	3

Question	Answer	Marks
2(a)(i)	<i>any two from:</i> trampling / dunes could collapse ; kills or disturbs, plants / animals; loss of habitat(s) ; loss of some species / extinctions; may drop litter; introduces invasive species; AVP;	2
2(a)(ii)	<i>any one from:</i> salt-tolerant; drought-tolerant / plants conserve or store water; wind-resistant; deep roots / wide roots; fast growing, plants / roots ;	1
2(a)(iii)	<i>any four from:</i> increase temperatures; leads to melting of ice caps / increased sea level; leads to flooding / sea covers sand dunes / dunes drown ; leads to (conditions that favour) invasive species extreme weather; leads to stronger winds / storms; leads to increased waves; erosion of sand; more salt on plants; greater competition (for resources); can't adapt fast enough / conditions are too extreme;	4
2(a)(iv)	compromised the ability of future generations to meet their own needs / didn't leave sand for future (generations) ;	1

Question	Answer	Marks
2(b)(i)	T AND V ticked;	1
2(b)(ii)	(scale is) 1 km = 0.4 (cm) OR 2 km = 0.8 (cm) OR (in cm X to nearest sand dune) = range 1.7 to 1.9 (cm) (distance is) range 4.2 to 5.0 (km);	2
2(b)(iii)	<i>any two from:</i> only women questioned / not a representative sample ; leads to biased results; too much data; will be expensive to conduct / analyse;	2
2(b)(iv)	<i>any one from:</i> easier / quicker, to collect or analyse or process answers; consistency of answer; limits possible answers;	1
2(b)(v)	<i>any two from:</i> employment; improvement in local or national economy; improvement in facilities / infrastructure;	2

Question	Answer	Marks
2(c)(i)	<i>any four from:</i> <i>use of transect and quadrat:</i> divide the transect, systematically / in equal distances / stated distances; defined size of quadrat, e.g. 25 cm × 25 cm, 50 cm × 50 cm, 1 m × 1 m; count the number of plant species, every stated distance / in quadrat; <i>recording and processing data:</i> record the results, in a table / using a tally; use a book to identify species; repeat and take a mean;	4
2(c)(ii)	<i>any three from:</i> the method only covers five (small) areas; five transects are not representative; lot of sand dune not sampled; plants not evenly distributed; longer transect needed (across whole area); AVP;	3

Question	Answer	Marks
3(a)(i)	sectors in clockwise rank order; largest first starting at 'noon'; correct plotting $\pm 4^\circ$; key completed and matches sector shading;	4
3(a)(ii)	<i>any two from:</i> cannot meet demand due to increased population; cost / shortage, of fossil fuels; lack, of renewables; cheaper to import than to generate;	2
3(a)(iii)	<i>any one from:</i> economic impact / cost; lack of energy security or described;	1
3(a)(iv)	<i>max [3] advantages or disadvantages</i> renewable; does not emit CO ₂ (at point of use); so does not contribute to climate change; does not emit SO ₂ / NO _x ; so does not contribute to acid rain; land around wind turbines can still be used; <i>disadvantages:;</i> not suitable if no wind or not strong / powerful enough; wind needs back up generation; turbines in area of natural beauty / visual / noise pollution;	4

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Question	Answer	Marks
3(b)(i)	<i>any four from:</i> buried in landfill takes up takes space; oil, leakage / spillage; toxic (to humans/animals); named toxin: benzene / lead / chromium / arsenic / dioxins / heavy metals; burning produces atmospheric pollution; named pollutant CO₂ / NO_x / VOCs, / SO₂ / hydrocarbons / CO / smoke / particulates; named effect on human health, e.g. cancer risk / damage liver / respiratory problems / skin irritation ; (oil) contaminates / pollutes, land / soil / water animals, suffocate / covered in oil or reduces development of animals; disrupts, food supply / food chain / ecosystems;	4
3(b)(ii)	900 (litres);	1
3(c)	<i>any four from:</i> reduce consumption; by turning off electrical devices; insulation of homes / name example; use energy efficient devices / vehicles; educate people on energy consumption; invest in / use more, renewables; increase taxes on fossil fuels; recycle to reduce energy use / stated example; generating electricity from waste products;	4

Question	Answer	Marks
4(a)	(plants in field) P / clover AND grass (best diet); sheep had greatest mass; OR (plants in field) Q / grass only (worst diet); sheep had least mass;	2
4(b)(i)	already had sufficient minerals in their diet (so not licking block) / did not use lick / already fully grown do not grow any bigger;	1
4(b)(ii)	<i>any one from:</i> no AND limited increase in mass /no effect on some sheep; yes AND all but two / most, sheep increased in mass; AVP;	1
4(b)(iii)	<i>any one from:</i> sheep, become ill / die / stop growing; mass, decreases / stops increasing ;	1
4(b)(iv)	select (two) sheep with the greatest mass; breed the sheep; choose offspring with the greatest mass; repeat / breed these offspring;	4
4(c)(i)	<i>any three from:</i> timber extraction / logging; roads; urbanisation / homes / industry; rock / mineral extraction; fuel;	3

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Question	Answer	Marks
4(c)(ii)	<i>any four from:</i> reduction in photosynthesis; (trees act as) carbon sink or store; fewer young or growing trees to remove / fewer mature trees store so more CO₂ or carbon ; change in decomposition (of leaves); change in carbon added to the ground; (deforestation leads to) increased carbon dioxide in atmosphere ; imbalance between respiration and photosynthesis; AVP;	4
4(d)(i)	<i>any four from:</i> populations dispersed / isolated areas; small areas can only support small populations; no safe corridors between populations; leads to limited breeding; limited numbers / limited gene pool; reduced habitat / deforestation; leads to limited food sources; (illegal) hunting; limited protected areas / protection laws not enforced; disease; climate change and stated impact, e.g. lack of food	4
4(d)(ii)	<i>any two from:</i> young less likely to die from disease / access to vaccination; other stated medical support; breeding support / increased gene pool; no threat from predators / hunting / poaching ; not released into the wild until mature / strong; no lack food / good food supply; AVP;	2